Predicting Suburban/Rural or Urban Background with Natural Language Processing & SVM

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Abstract

This project uses survey responses to study how individuals from suburban/rural and urban backgrounds view their hometowns and childhood experiences. Responses from a 3-question survey were analyzed using Natural Language Processing (NLP) to see if patterns in textual data were sufficient enough to predict a participant's upbringing environment. The dataset, which had responses from 93 participants, was used to train a Support Vector Machine (SVM) model. This SVM model was used to develop an interactive prediction tool that attempted to classify a person's background into one of two groups: Suburban/Rural or Urban. The study highlights differences in trends in both categories and shows that NLP can effectively predict (but not perfectly) whether someone grew up in a Suburban/Rural or environment.





Introduction & Significance

People make assumptions about each other's childhood experiences based on where they grew up.

People who grew up in suburban/rural areas tend to think their childhoods were very different from those of people who grew up in big cities and vice versa.

Research Questions:

1. Are people's childhood experiences in their hometowns really that different based on whether they came from a suburban/rural or urban background?
2. Can a Machine Learning model accurately predict whether someone grew up in a city or a suburb/rural area based on how they describe their hometown and childhood?

I designed a short survey to gather information about people's perceptions about their own hometowns and childhoods. I then used NLP Text Preprocessing techniques to clean the textual data and transform words into numerical features.

A Support Vector Machine model used these numerical features to create an interactive predictive model.

Methods

Part 1. DATA COLLECTION

- A brief 3-question survey was designed and sent to participants.
- The survey received responses from 93 participants (n=93).
- 44 participants grew up in an urban background (~47%)
- 49 participants grew up in a suburban/rural background (~53%)
- Survey questions:

1. Did you primarily grow up in an urban or suburban/rural setting? (Multiple choice)
2. What are some good things and some bad things about where you grew up? (short answer or paragraph response)
3. What do you remember most about your childhood? (Short answer or paragraph response)

Part 2. TEXT PREPROCESSING

- Each participant's answers to Questions 2 and 3 were combined for textual analysis
- TF-IDF vectorization was used to convert textual data into numerical features for SVM
- Common English stopwords were removed

Part 3. SVM Model

- class_weight="balanced" was used to account for possible class imbalance
- Pipeline with TF-IDF vectorization and SVM model was built

Part 4. HYPERPARAMETER TUNING

- GridSearchCV was used to locate best parameters:
  - Maximum words used: 1000
  - Words appearing in less than 1 document were not used
  - Unigrams, 2-word phrases, 3-word phrases & 4-word phrases were analyzed

Part 5. MODEL EVALUATION & METRICS

- SVM model was trained on 4 folds of full dataset (around 80%) and tested on 1 fold (around 20%) in each iteration
- Created cross-validated predictions across all 5 folds
- Evaluated model based on macro average F1 score (calculated based on combined predictions from all 5 validation folds)

Part 6. INTERACTIVE PREDICTION MODEL

- Designed function that asks user 2 questions
  - What are some good things and some bad things about where you grew up?
  - What do you remember most about your childhood?
- Pipeline analyzed input text and used SVM model to predict whether user grew up in an "Urban" or "Suburban/Rural" hometown

Part 7. HUGGING FACE EMOTION DETECTION & HEATMAP VISUALIZATION

- Classified each participants' overall responses based on dominant emotion detected
- Displayed percentages (not counts) of emotions detected in "Urban" and "Suburban/Rural" classes in a heatmap visualization

Part 8. THEMATIC EXTRACTION WITH LDA & WORDCLOUDS

- Added custom stopwords to attempt to find better patterns in themes and created one WordCloud per class to show top themes for both groups

Results

```
Classification Report:
              precision    recall  f1-score   support

           0       0.74      0.80      0.76        49
           1       0.75      0.68      0.71        44

    accuracy                           0.74        93
   macro avg       0.74      0.74      0.74        93
weighted avg       0.74      0.74      0.74        93
```

Figure 1. SVM Model Performance Metrics (0=Suburban/Rural, 1=Urban

```
--- Did you grow up in an urban or suburban/rural background? ---
What are some good things and some bad things about where you grew up?
We lived in a big house in a peaceful neighborhood. There weren't a lot
of places in town to hangout in.
What do you remember most about your childhood?
I remember playing outside in a park!

 Prediction: Suburban/Rural
```

```
--- Did you grow up in an urban or suburban/rural background? ---
What are some good things and some bad things about where you grew up?
I liked how there were so many different types of food to try! I didn't
like how small our apartment was.
What do you remember most about your childhood?
I remember my parents taking me to try different restaurants.

 Prediction: Urban
```

Figure 2 & 3. example of predictions from interactive model

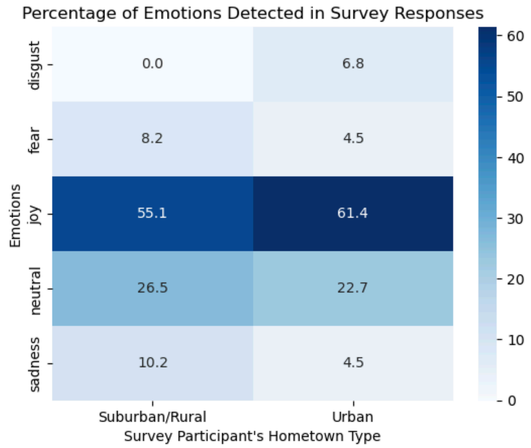


Figure 4. Heatmap Visualization of Percentage of Emotions Detected in both groups' responses



Figure 5 & 6. Word cloud for Top Themes in Suburban/Rural & Urban Responses

Discussion

- The SVM model produced a macro average F1-score of 0.74, which is decent but not perfect.
- The Urban category had a higher percentage of joyful responses (61.4%) than the Suburban/Rural category (55.1%).
- Disgust was the most common negative emotion that was detected in the Urban category's responses.
- Sadness was the most common negative emotion that was detected in the Suburban/Rural category's responses.
- The model was around 5% better at predicting if someone came from a Suburban/rural background than an Urban one. This could be due to more detectable patterns or a slight class imbalance.
- This dataset had 49 samples from the suburban/rural class and 44 samples from the urban class, so the proportion is pretty balanced even though it's not perfect. However, the SVM model was trained to account for potential class imbalance.
- The dataset was not a stratified sample and only contained 93 total samples.
- Participants from suburban and rural backgrounds might have very different experiences, but due to the small dataset, these were combined into one group. Some participants might have grown up in both environments.
- Additional custom stopwords were not added to the SVM/prediction model, but they were added to the WordCloud visualizations in an attempt to improve thematic extraction of keywords.

Conclusion

- This project aimed to use NLP techniques to see if open-ended survey responses could be used to accurately predict whether someone grew up in an urban or suburban/rural area.
- The Linear SVM model reached a macro average F1 score of 0.74 which demonstrates that there were enough sufficient differences between each class to make pretty accurate (but not perfectly accurate) predictions.
- Notable limitations included the dataset not being a stratified sample and the number of participants being small (n=93).

References

OpenAI. (2026). *ChatGPT* [Large language model]

Sarkar, D. (2019). Text Analytics with Python: A practitioner's guide to natural language processing. Apress.

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